

Ira Russell Check

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TECHNICAL SKILLS

Languages: C#, Python, Java, C, JavaScript, HTML, CSS

Web & Frameworks: React, Node.js, Vite, TailwindCSS, Firebase, .NET, WPF, XAML

Other Tools & Platforms: Unity, OpenCV, Arduino IDE, Git, GitHub, Visual Studio, Rider, IntelliJ, CLion

Core Concepts: Object-Oriented Programming, Data Structures and Algorithms, Unit Testing, JSON Serialization, Serial Communication, Software Development Lifecycle

WORK EXPERIENCE

Web Developer | August 2026 - Present

The Broken Spoke Bicycle Shop, Easton, PA (Remote)

- Redesigning and rebuilding a small business's production website using React, Vite, and TailwindCSS, replacing an outdated legacy site.
- Design and develop a custom content management system on Firebase, giving the business owner independent, no-code control over website content and eliminating reliance on a developer for routine updates.
- Gather and translate business requirements into technical specifications, balancing usability, maintainability, and budget constraints in a full-stack development role.

Back of House Team Member | May 2026 - August 2026

Chick-fil-A, State College, PA

Associate | May 2024 - September 2025

Giant Food Stores, Walnutport, PA

PROJECTS

Checkers-Playing Robotic Arm | March 2026 - Present

Python, C, OpenCV, Arduino

- Build an autonomous checkers-playing robotic system that integrates a custom game engine, computer vision, AI decision-making, and real-time hardware control.
- Develop a complete checkers game engine in Python featuring move validation, board-state management, and a Minimax algorithm with alpha-beta pruning for automated move selection.
- Engineer a computer vision pipeline using OpenCV to convert live camera input into accurate, machine-readable board state data for autonomous gameplay.
- Design a custom serial communication protocol between the Python control system and an Arduino motor controller, incorporating error detection to identify and recover from failed hardware commands.

Modular Agent-Based Simulation Engine | June 2025 - January 2026

C#, ScriptHookV / ScriptHookVDotNet

- Architected and built an extensible agent-based simulation framework in C# supporting 100+ concurrent NPC agents in dynamic, user-defined environments, achieving 450+ public downloads.
- Implemented an event-driven software architecture with support for external code extensions and human-readable INI-based configuration, enabling non-technical users to script custom NPC behavior without writing code.
- Profiled and optimized runtime performance to under 2ms per simulation tick, reduced latency spikes above 5ms to under 0.1% of ticks, and kept memory usage under 60MB on test hardware.

Input Remapper Tool | February 2025

C#, WPF, XAML, XInput, Windows API

- Developed a Windows desktop accessibility utility that translates Xbox controller input into low-level mouse and keyboard events using the XInput and Win32 APIs.
- Designed and implemented a responsive graphical user interface in WPF and XAML for configuring custom controller-to-input mappings.
- Engineered custom input polling logic that reduced application CPU usage by 20 percent.

EDUCATION

B.S. in Computer Science | Expected: May 2027

Pennsylvania State University, University Park, PA - GPA: 3.26